

Trustworthiness, Truthfulness and Honesty

Introduction

Trustworthiness, truthfulness, and honesty are basic virtues of a professional engineer. Engineering work affects public safety, company decisions, research results, and society, so people must be able to trust engineers.

According to Fleddermann, engineering ethics includes issues like fraud, fairness, honesty in research and testing, environmental protection, and conflicts of interest. This shows that honesty is not only a personal quality; it is a professional duty.

1. Trustworthiness

Trustworthiness means being reliable, responsible, and worthy of confidence. A trustworthy engineer gives correct information, keeps promises, protects confidential data, and does not misuse professional power.

In engineering, trust is very important because employers, clients, government, and the public usually cannot check every technical detail themselves. They depend on the engineer's professional judgment.

Fleddermann connects trustworthiness with virtue ethics. Virtue ethics focuses on good character traits such as responsibility, honesty, competence, loyalty, fairness, caring, respect, and trustworthiness.

Example:

If an engineer signs a safety report, people trust that the report is based on proper testing and real data.

2. Truthfulness

Truthfulness means saying what is true and not hiding important facts. An engineer must be truthful in reports, public statements, test results, research papers, and expert opinions.

Engineering codes strongly support truthfulness. The NSPE and ASCE codes state that engineers should issue public statements only in an **objective and truthful** manner. Engineers must include relevant information and should not make untrue, unfair, or exaggerated claims.

Example:

If a product fails a safety test, the engineer must report the failure honestly instead of hiding it to protect the company.

3. Honesty

Honesty means avoiding lies, fraud, cheating, deception, and manipulation of facts. An honest engineer does not falsify data, copy work, accept bribes, or mislead clients and the public.

Fleddermann explains honesty as an important virtue in professional life. He also states that honesty and trustworthiness are important virtues for engineers and are reflected in engineering codes of ethics.

Honesty is required in:

- research and experiments
 - testing and inspection
 - reports and documentation
 - public statements
 - cost estimation and bidding
 - use of confidential information
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4. Truthfulness Responsibility in Research

Truthfulness is especially important in research. Research is supposed to be unbiased, and results should be reported truthfully. A researcher must not change data just to support the desired conclusion.

Fleddermann explains that researchers are responsible for performing experiments objectively. In research, dishonesty can damage the reputation of students, scientists, universities, and the profession.

Example:

If Alice fabricates experimental data to match the desired result, she violates truthfulness responsibility because she is not reporting what actually happened.

5. Relation with Professional Codes

Professional codes make honesty and truthfulness official duties of engineers. These codes say engineers must be faithful agents of employers and clients, avoid deceptive acts, and conduct themselves honorably and responsibly.

The ASCE code also says engineers must uphold the honor, integrity, and dignity of the profession and should have zero tolerance for bribery, fraud, and corruption.

This means an engineer should not think, “Everyone does it.” Even if dishonesty benefits the company, it damages public trust and the engineering profession.

6. Importance in Engineering

Trustworthiness, truthfulness, and honesty are important because engineering decisions can affect lives. If an engineer lies in a report, hides a defect, or changes data, people may be injured or killed.

These virtues also protect the reputation of the profession. Once the public loses trust in engineers, even good engineering work becomes doubtful.

Therefore, an engineer must be honest even under pressure from management, clients, deadlines, or financial problems.

Conclusion

Trustworthiness, truthfulness, and honesty are essential qualities of a good engineer. A trustworthy engineer is reliable, a truthful engineer reports facts correctly, and an honest engineer avoids fraud and deception.

Engineers must be objective in reports, truthful in public statements, honest in research, and fair in professional dealings. These virtues protect public safety, professional dignity, and the trust society places in engineers.

How to Apply in Case Studies

In a case study, first identify where dishonesty happened. It may be false data, hidden defects, copied code, fake testing, wrong reporting, or misleading statements.

Then explain which duty is violated:

- **Trustworthiness:** people can no longer rely on the engineer
- **Truthfulness:** facts were hidden or misrepresented
- **Honesty:** deception, fraud, or manipulation was done

Finally, give the ethical decision. The engineer should correct the record, report facts honestly, refuse to falsify data, and protect public safety.

Short Case Study: Alice and Fabricated Data

Alice manipulates experimental data in a research project so that the results match the desired outcome.

This is unethical because she violates truthfulness responsibility. Research must be objective and results must be reported truthfully. Her false data can mislead the team, journal, future researchers, and industries that may use the material.

The correct action is to stop publication, repeat the experiments honestly, correct the data, and report the misconduct through proper academic procedure.

Memorization Checklist

Remember:

T-T-H

T — Trustworthiness: Be reliable and responsible.

T — Truthfulness: Report facts objectively and completely.

H — Honesty: Avoid fraud, lies, deception, and data manipulation.

5-Mark Exam Answer

Trustworthiness, truthfulness, and honesty are basic virtues of a professional engineer. Trustworthiness means being reliable and worthy of confidence. Since the public, employers, and clients cannot check every technical detail, they depend on the engineer's judgment. A trustworthy engineer gives correct information, keeps promises, and does not misuse professional power.

Truthfulness means reporting facts correctly and not hiding important information. Engineering codes require engineers to issue public statements only in an objective and truthful manner. Engineers should not make false, unfair, or exaggerated claims. Honesty means avoiding lies, fraud, cheating, bribery, and manipulation of data.

These values are very important in research and testing. Research should be unbiased and results should be reported truthfully. If an engineer fabricates data, hides failed tests, or falsifies reports, public safety and professional trust are damaged. Therefore, engineers must remain honest in reports, experiments, public statements, and professional dealings, even under pressure from employers or clients.